Autism Visual Supports

Printable, or tappable. Same board.



OUTSYSTEMS ODC · AGENTIC AI · TWILIO

Build for the Future Hackathon 2026

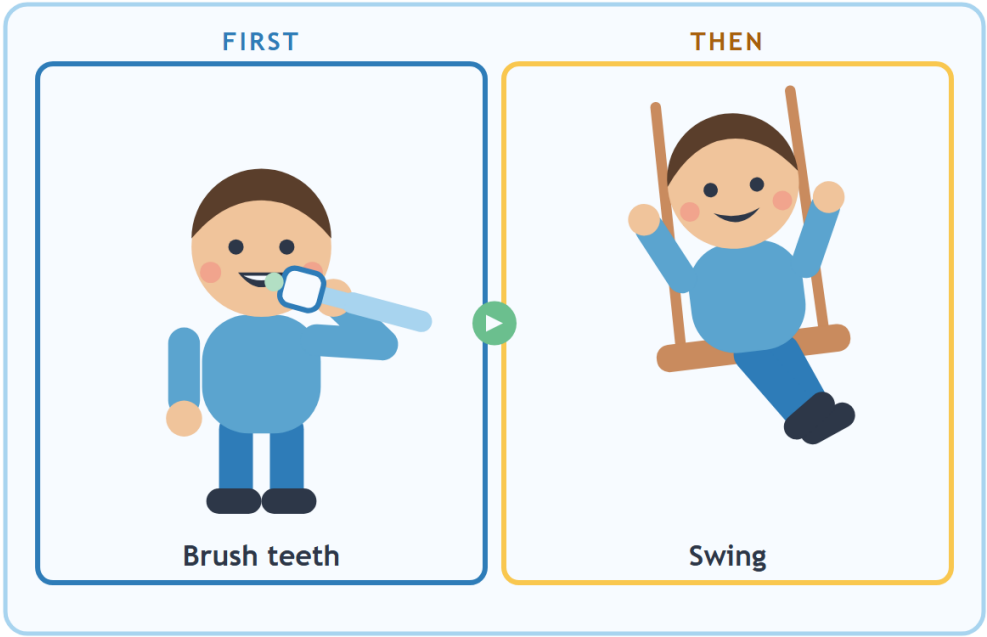
A.J. Aronoff

Four minutes to get out the door

A child is on the floor with a toy. In four minutes they need shoes on with their go-bag and they need to be in the car.

“We have to go now” does not work. Saying it louder works less.

What works is a picture.



Two cards, in order, where the child can see them.

First shoes. Then swing.

The method works. The making does not

What the research says

First–Then boards, visual schedules and parameter cards are documented practice in the Vanderbilt TRIAD toolkit, published through the Autism Speaks Autism Treatment Network.

And in the standards

The National Autism Center lists Schedules, Antecedent Packages and Story-based Intervention among the interventions with compelling scientific evidence.

So why do most parents make two or three boards, ever?

- Find pictures the child recognises
- Cut them to size
- Keep every label to three words or fewer
- Laminate, then add Velcro

40 minutes and the morning you needed it was this morning

“You must give them the reinforcing activity or item after they complete the ‘first’ task. Otherwise, your child may not trust the board the next time you use it.”

Vanderbilt Kennedy Center TRIAD / Autism Speaks ATN — Visual Supports and Autism (2011)

A board the child has stopped trusting is worse than no board. The next one starts from suspicion.

Three questions in. Three things out

The child’s profile — age, reading level, photos or icons, special interests, sensory triggers — is filled in once and reused forever. Every later board is three questions: which routine, what is hard about it, what the child actually wants afterwards.

1

The cards

Three words maximum. Phrased positively — “Walk in hallway”, never “Don’t run”. Printable with cut lines for lamination, or full-screen on a tablet.

2

The caregiver script

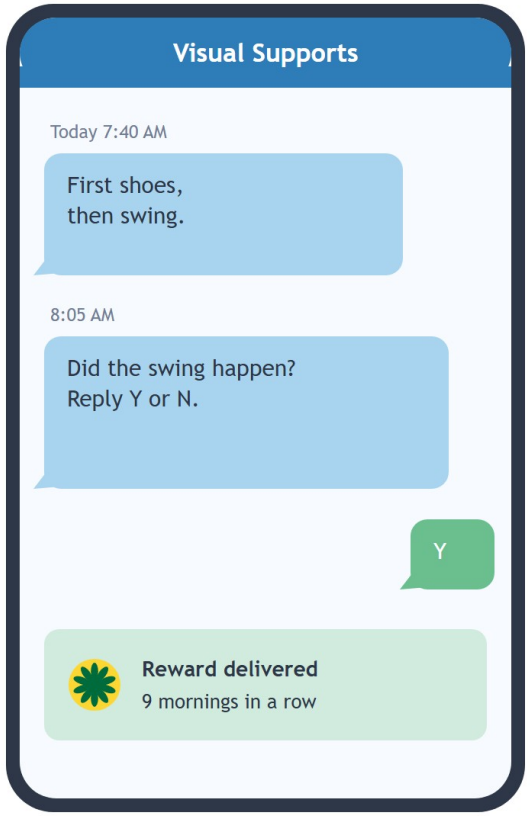
The one or two low-demand phrases to say out loud. “First shoes, then swing.” Then “All done shoes. Now swing!” Most tools stop at the picture. The words are what parents fumble.

3

The implementation plan

Where to put the board, the prompt hierarchy from least to most intrusive, and what to do when it goes wrong.

The loop nobody else closes



At the moment the routine runs, the parent is at the front door holding a pair of shoes. They are not holding a tablet.

Twilio carries the script

The caregiver phrase arrives by SMS when the routine is due — the only channel that reaches a parent mid-morning.

And asks one question after

“Did the swing happen? Reply Y or N.” One character writes the field the whole method depends on.

The home screen shows how reliably the promise has been kept — not how many boards you have made.

Next: social stories

Not in the seven-day build — but it is the next support type, and it costs one branch rather than a second app. A First–Then board handles the transition happening now; a social narrative handles what has not happened yet: the dentist, the fire drill, the new classroom.

Same three-part output

Pictures, the words to say, and the plan — written in the first person, delivered before the event rather than during it.

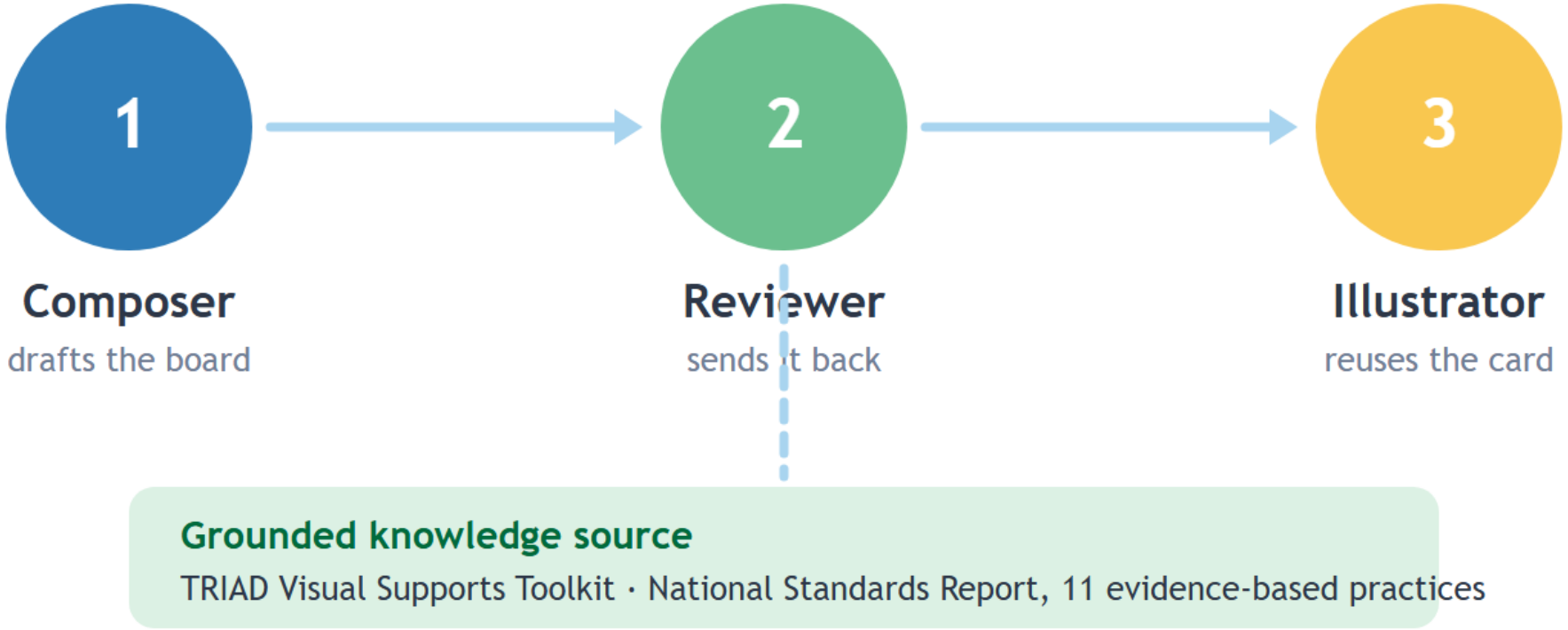
Same evidence base

Story-based Intervention: who, what, when, where and why, answered ahead of time.



Read in the shade, by the light of the cloud that wrote it.

Agentic AI, not a single prompt



Each agent has a job small enough to inspect, and Agent Workbench exposes the reasoning path — which makes it demonstrable in five minutes.

Guardrails live in the app, not the prompt

Three words maximum

Rejected and regenerated. The toolkit’s limit, not a style preference.

Positive phrasing only

“Don’t run” is refused in favour of “Walk”.

No Then card, no save

A First–Then board cannot exist without its reward.

No diagnosis, no clinical advice

Self-injury or escalation in the text surfaces referral guidance instead of a board.

The child’s name never leaves

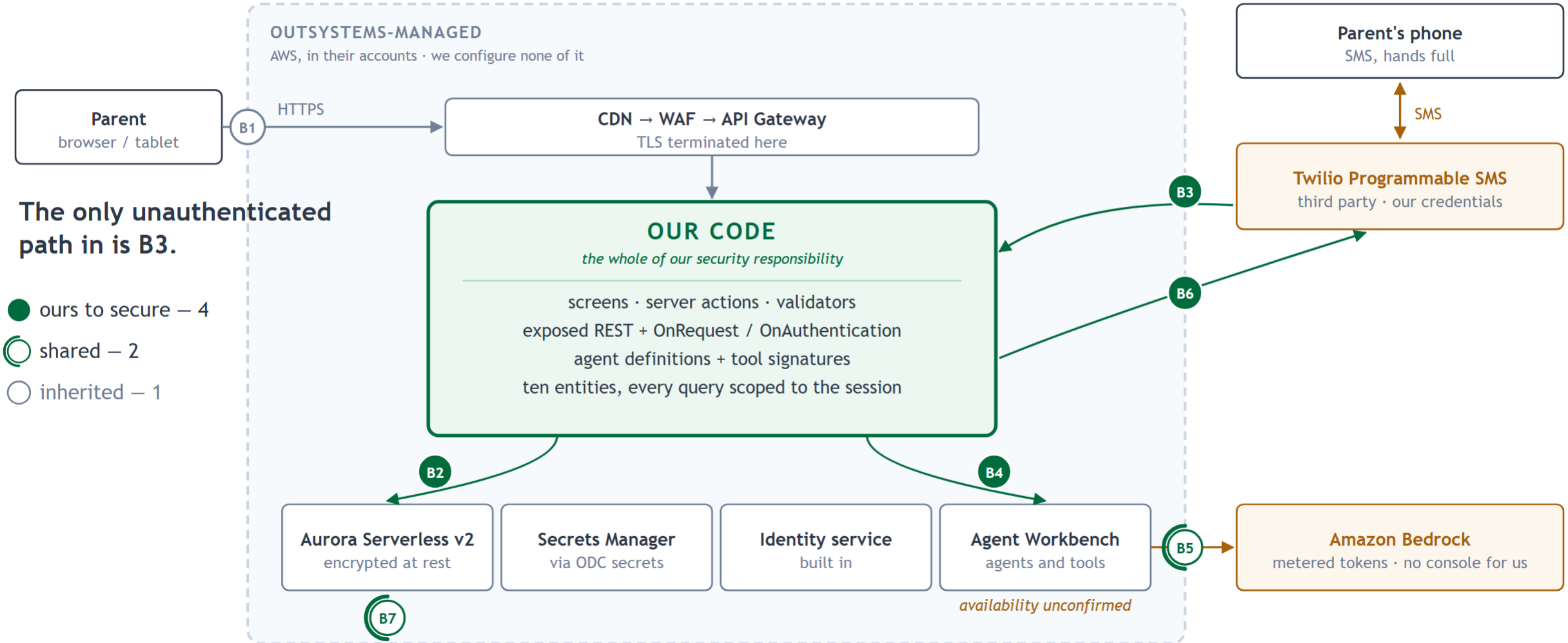
Image prompts say “a child”. Profiles are never training input.

Open source

Per the hackathon rules, so any family or nonprofit can run it.

Trust boundaries

Autism Visual Supports on OutSystems Developer Cloud · who owns which control



Seven days, demo path first

2 Nov	**Kickoff**	Re-plan against the ODC features and scoring announced there.
3–4 Nov	**Model + agents**	Entities, child profile, Composer and Reviewer on the First–Then branch.
4 Nov	**Mid submission**	Problem, use case, AI services and integrations.
5–7 Nov	**The demo path**	Wizard, editor, card library, print sheet, tablet mode, Twilio loop.
8 Nov	**UX pass**	Every screen the video touches. Nothing new built.
9 Nov	**Submit**	Five-minute walkthrough the judges can replicate. 11:59 PM CET.

ODC access opens only after registration closes on 28 October — so nothing can be built before the 2 November kickoff.

Why this team

My daughter is autistic. My wife will be the first test user.

I have been making these boards by hand for years, and I know exactly which forty minutes this app gives back.



Nonprofit partner: Autism Speaks — whose Autism Treatment Network published the toolkit this is built on.